

LEXICON LEARNING ANALYTICS & AI FEEDBACK PLATFORM

Project Summary & Internship Roadmap

From Learning Data to Actionable Student Feedback



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From Learning Data to Actionable Student Feedback



INTERNSHIP PERIOD

03 August 2026 –
06 October 2026
(9 Weeks)



LOCATION

Lexicon Syd,
Malmö, Sweden



INTERN

Ayşe Rana Karakuyu



PROJECT GOAL

Build a smart platform to manage student data, attendance, assessments and provide AI-powered feedback to support student growth.

KEY FEATURES



User & Role Management

Admin, Teacher, Student



Course & Group Management

Create courses, groups and assign teachers



Student Import

Manual, CSV/Excel, PDF, DOCX (AI-assisted)



Attendance Management

Morning 09:00–12:00
Afternoon 13:00–16:00



Assessments & Quizzes

Projects, Assignments, Quizzes, Rubrics



Analytics & Dashboards

Progress, Trends, Competency Matrix



AI Feedback Engine

AI insights, Teacher review, Recommendations



PDF Reports & Certificates

Complete student progress reports

HOW IT WORKS



1. COLLECT

Data from attendance, assessments, quizzes, projects and activities



2. ASSESS

Evaluate performance using rubrics and scores



3. ANALYZE

Generate statistics, charts and competency analysis



4. UNDERSTAND

Identify strengths, weaknesses and patterns



5. RECOMMEND

AI suggests feedback and learning recommendations



6. REPORT

Teacher reviews and approves. PDF reports for student

9-WEEK INTERNSHIP ROADMAP

WEEK 1 03–07 AUG



Analysis & Planning

- Understand requirements
- Define scope
- User stories
- Project backlog

Deliverable:
Requirements & Project Plan

WEEK 2 10–14 AUG



Design

- Architecture design
- UML diagrams
- Database design
- Wireframes

Deliverable:
Design Docs & Wireframes

WEEKS 3–4 17–28 AUG



Core Development (Setup & Foundation)

- Environment setup
- Authentication & roles
- Courses & groups
- Student management
- Basic UI & APIs

Deliverable:
Working Core Modules

WEEKS 5–6 31 AUG–11 SEP



Features Development

- Attendance (09:00–12:00 & 13:00–16:00)
- Projects & Assessments
- Quizzes
- Rubrics & Scoring

Deliverable:
Feature Complete (MVP Functional)

WEEKS 7–8 14–25 SEP



Analytics & AI Integration

- Dashboards & charts
- Competency Matrix
- AI Feedback Engine
- Teacher review flow
- PDF Report

Deliverable:
AI Feedback & Reports

WEEK 9 28 SEP–06 OCT



Finalization

- Testing & bug fixing
- UI/UX improvements
- Deployment
- Documentation
- Final presentation

Deliverable:
Deployed MVP & Final Presentation

SUGGESTED TECHNOLOGY STACK



Frontend

React / Next.js, TypeScript, Tailwind CSS



Backend

ASP.NET Core Web API (or Node.js / Express)



Database

PostgreSQL / MySQL



Authentication

JWT / ASP.NET Core Identity / Auth0



AI Integration

OpenAI API (or similar LLM API)



DevOps & Tools

GitHub, Docker (optional), CI/CD, Azure / Vercel

KEY OUTPUTS



Working MVP



GitHub Repository



AI-Powered Feedback



Analytics & Dashboards



PDF Student Reports



Technical Documentation



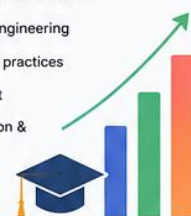
Final Presentation



Deployed Application

EXPECTED OUTCOMES

- ✓ Full-stack web development experience
- ✓ Real-world data analysis & visualization
- ✓ AI integration & prompt engineering
- ✓ Software testing & quality practices
- ✓ Agile project management
- ✓ Professional documentation & presentation
- ✓ Portfolio-ready project



This project transforms learning data into insights, personalized feedback and student success.



Learn. Build. Analyze. Improve. Succeed.
Let's build the future of learning together!

1. Project in One Page

Project Goal

Build a practical web platform that collects student learning data and turns it into clear progress analytics, teacher feedback, AI-assisted recommendations, and professional PDF reports.

What the system will manage

- Students, teachers, courses and groups
- Daily attendance: Morning 09:00–12:00 and Afternoon 13:00–16:00
- Projects, assignments, quizzes and assessments
- Competency / skill development
- Certificates and learning achievements
- Teacher comments and AI-assisted feedback
- Charts, dashboards and student progress reports

Simple system flow



Who uses it?

ADMIN Manage teachers, courses, groups and students	TEACHER Attendance, assessment, feedback and analytics	STUDENT Progress, results, feedback and recommendations
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2. Core Modules

MODULE	PURPOSE
User & Role Management	Admin, Teacher and Student roles with role-based access.
Course & Group Management	Create courses/groups, assign teachers and manage membership.
Student Management & Import	Manual entry and CSV/Excel for MVP; PDF/DOCX + AI extraction as advanced.
Attendance	09:00–12:00 morning and 13:00–16:00 afternoon; Present/Late/Absent/Excused.
Assessments	Projects, assignments, quizzes, presentations and final projects.
Rubrics & Competencies	Score projects by criteria and build a competency matrix.
Learning Analytics	Progress trends, group statistics, charts and dashboards.
AI Feedback Engine	Generate draft feedback from attendance, results, competencies and teacher comments.
PDF Student Report	Professional report with scores, charts, feedback and recommendations.
Certificates	Store certificates and learning achievements.

Signature feature: AI feedback

Recommended workflow

Student Data → AI Analysis → AI Draft Feedback → Teacher Review/Edit → Approved Feedback → Student

Principle: AI supports the teacher; it does not replace professional judgment.

3. Planning, Design & Technology

Before coding

- Define project scope and MVP
- Identify users, roles and use cases
- Write functional and non-functional requirements
- Create user stories and product backlog
- Design system architecture
- Create UML: Use Case, Class and Sequence; Activity/Component when useful
- Design database and ER diagram
- Create user flows and wireframes
- Select and justify the technology stack

Recommended technology stack

AREA	RECOMMENDED	ALTERNATIVES
Frontend	React / Next.js + TypeScript	Tailwind CSS
Backend	ASP.NET Core Web API	Node.js / Express
Database	PostgreSQL / SQL Server / MySQL	Other relational DB
Authentication	ASP.NET Core Identity / JWT	Auth0 / Clerk
AI	OpenAI API	Other suitable LLM API
Charts	Chart.js / Recharts	ApexCharts / equivalent
API	Swagger / OpenAPI + Postman	Equivalent tools
Planning	GitHub Projects / Jira / Trello	Notion
Design	Figma + Draw.io	Excalidraw / StarUML
DevOps	GitHub + Azure/Vercel	Docker / GitHub Actions

Responsible AI-assisted development

- Use AI for research, brainstorming, coding, refactoring, testing and documentation.
- Validate and understand AI-generated output before implementation.
- Keep a simple record of meaningful AI usage.

4. 9-Week Internship Roadmap

PERIOD	PHASE	MAIN OUTPUT
Week 1 03–07 Aug	Analysis & Planning	Scope, requirements, users, user stories, backlog
Week 2 10–14 Aug	Architecture & Design	Architecture, UML, ERD, wireframes, technology selection
Weeks 3–4 17–28 Aug	Core Development	Setup, authentication, roles, courses, groups, APIs
Weeks 5–6 31 Aug–11 Sep	Feature Development	Students, attendance, assessments, quizzes, rubrics
Weeks 7–8 14–25 Sep	Analytics & AI	Dashboards, competency matrix, AI feedback, recommendations, PDF
Week 9 28 Sep–02 Oct	Finalization	Testing, security, deployment, documentation, presentation

MVP priority

MUST HAVE

Authentication & roles • Courses/groups • Students • Attendance • Assessments • Quizzes • Student dashboard • Competency matrix • AI feedback • PDF report • GitHub • Deployment

SHOULD HAVE: CSV/Excel import, group analytics, AI learning recommendations, feedback templates, certificates, course timeline.

NICE TO HAVE: PDF/DOCX import, AI document extraction, risk indicator, notifications, audit log, Docker, CI/CD, advanced AI/RAG.

Final deliverables

- Working full-stack MVP
- GitHub repository
- Architecture, UML and database documentation
- API and testing documentation
- AI integration documentation
- Professional PDF student report
- Live deployment
- Final presentation and internship report

5. Learning Outcomes & Success

LEARNING AREA	EXPECTED EXPERIENCE
Software Engineering	Requirements analysis, architecture, UML, Agile planning, Git workflows
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Full-Stack Development	Modern frontend, REST APIs, backend, database integration, authentication
Data & Analytics	Data modelling, charts, progress analytics, competency analysis
AI	LLM/API integration, prompt engineering, AI-assisted development, human-in-the-loop feedback
Quality & Security	Unit/API/integration testing, debugging, validation, role-based security
Professional Skills	Documentation, project ownership, communication, presentation, problem solving

Success criteria

- Functional and deployable MVP delivered within the internship.
- Complete flow demonstrated: data collection → assessment → analytics → AI feedback → reporting.
- Trainee can explain architecture, database, APIs, testing and technology choices.
- AI feedback is reviewed and validated by the teacher.
- Documentation is sufficient for another developer to understand and continue the project.

Project Vision

Transform learning data into actionable insights, personalized feedback and better student development — while giving the trainee real-world experience across the complete software development lifecycle.